YDLIDAR/ydlidar_ros2_driver: ydlidar driver package under ros2

YDLIDAR ROS2 Driver

ydlidar_ros2_driver is a new ros package, which is designed to gradually become the standard driver package for ydlidar devices in the ros2 environment.

How to install ROS2

ubuntu

windows

How to Create a ROS2 workspace

Create a workspace

Build & Install YDLidar SDK

ydlidar_ros2_driver depends on YDLidar-SDK library. If you have never installed YDLidar-SDK library or it is out of date, you must first install YDLidar-SDK library. If you have installed the latest version of YDLidar-SDK, skip this step and go to the next step.

1. Download or clone the YDLIDAR/YDLidar-SDK repository on GitHub.
2. Compile and install the YDLidar-SDK under the *build* directory following README.md of YDLIDAR/YDLidar-SDK.

Build ydlidar_ros2_driver

1. Clone ydlidar_ros2_driver master branch from github for old version:

```
git clone https://github.com/YDLIDAR/ydlidar_ros2_driver.git
ydlidar_ros2_ws/src/ydlidar_ros2_driver
```

Clone ydlidar_ros2_driver humble branch from github for humble,jazzy,etc:

```
git clone -b humble https://github.com/YDLIDAR/ydlidar_ros2_driver.git
ydlidar_ros2_ws/src/ydlidar_ros2_driver
```

2. Build ydlidar_ros2_driver package :

```
cd ydlidar_ros2_ws
colcon build --symlink-install
```

Note: install colcon see

```
hwu@hwu-Nucs:~/ros2_ws$ colcon build --symlink-install
Starting >>> ydlidar_ros2_driver
Finished <<< ydlidar_ros2_driver [6.11s]
Starting >>> ydlidar_ros2_driver
Finished <<< ydlidar_ros2_driver [8.25s]
Summary: 2 packages finished [8.25s]
hwu@hwu-Nucs:~/ros2_ws$

>>>ydlidar_ros2_driver
--- stderr: ydlidar_ros2_driver
CMake Error at CMakeLists.txt:15 (find_package):
  Are you trying to "find_package(ydlidar_sdk CMake) in CMAKE_MODULE_PATH this project
  has asked CMake to find a package configuration file provided by
  "ydlidar_sdk", but CMake did not find one.

  Could not find a package configuration file provided by "ydlidar_sdk" with
  any of the following names:

    ydlidar_sdkConfig.cmake
    ydlidar_sdk-config.cmake

  Add the installation prefix of "ydlidar_sdk" to CMAKE_PREFIX_PATH or set
  "ydlidar_sdk_DIR" to a directory containing one of the above files.  If
  "ydlidar_sdk" provides a separate development package or SDK, be sure it
  has been installed.


Failed   <<< ydlidar_ros2_driver          [ Exited with code 1 ]
Aborted  <<< ydlidar_ros2_driver[0.8s]

Summary: 0 packages finished [1.43s]
  1 package failed: ydlidar_ros2_driver
  1 package aborted: ydlidar_ros2_driver
  1 package had stderr output: ydlidar_ros2_driver
hwu@hwu-Nucs:~/ros2_ws$
```

3. Package environment setup :

```
source ./install/setup.bash
```

Note: Add permanent workspace environment variables. It's convenient!If the ROS2 environment variables are automatically added to your bash session every time a new shell is launched:

```
echo "source ~/ydlidar_ros2_ws/install/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

4. Confirmation To confirm that your package path has been set, printenv the grep -i ROS variable.

```
printenv | grep -i ROS
```

You should see something similar to:`OLDPWD=/home/tony/ydlidar_ros2_ws/install`

5. Create serial port Alias [optional]

```
chmod 0777 src/ydlidar_ros2_driver/startup/*
sudo sh src/ydlidar_ros2_driver/startup/initrules.sh
```

Note: After completing the previous operation, replug the LiDAR again.

Configure LIDAR Default parameter file

```
ydlidar_ros2_driver_node:
  ros__parameters:
    port: /dev/ttyUSB0
    frame_id: laser_frame
    ignore_array: ""
    baudrate: 230400
    lidar_type: 1
    device_type: 0
    isSingleChannel: false
    intensity: false
    intensity_bit: 8
    sample_rate: 9
    abnormal_check_count: 4
    fixed_resolution: true
    reversion: false
    inverted: false
    auto_reconnect: true
    support_motor_dtr: false
    angle_max: 180.0
    angle_min: -180.0
    range_max: 64.0
    range_min: 0.01
    frequency: 10.0
    invalid_range_is_inf: false
    debug: false
```

Note: It needs to be modified according to LiDAR actual situation.

Run ydlidar_ros2_driver

Run ydlidar_ros2_driver using launch file

The command format is :

```
ros2 launch ydlidar_ros2_driver [launch file].py
```

1. Connect LiDAR uint(s).

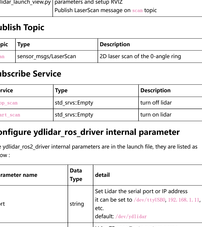
```
ros2 launch ydlidar_ros2_driver ydlidar_launch.py
```

or

```
launch $(ros2 pkg prefix
ydlidar_ros2_driver)/share/ydlidar_ros2_driver/launch/ydlidar.py
```

2. RVIZ

```
ros2 launch ydlidar_ros2_driver ydlidar_launch_view.py
```


RviZ - pose_grid
Displays
Fixed Frame
Global Options
Grid
Reference Frame
Value
Plane
XY
Plane Cell Count
10
Color
160; 160; 164
Enabled
Cell Size
1
Normal Cell Count
10
View Controller
Current View
Type
Orbit (rviz_default_plugins)
Zoom
3
Distance
10.3
Focal Point
X
0.0
Y
0.0
Z
0.0
Save
Remove
Rename
LaserScan
Topic
/scan
Size (Pixels)
3
Style
Flat Squares
Decay Time
0
Position Transformation
XYZ
Color Transformation
Intensity
Channel #0
intensity
Selectable
Use Rainbow
Invert Rainbow
Min Color
255; 255; 255
Max Color
255; 255; 255
Min Intensity
0
Max Intensity
4096
Autocompute Intensity Bounds
Queue Size (Messages)
10
Reliability Policy
Best Effort
Durability Policy
Volatile
Filter size
0
Use Fixed Frame
Use Clock
Frames
All enabled
Hide All
Show All
Expand All
Collapse All
LaserScan
Topic: /scan
Type: sensor_msgs/msg/LaserScan
Publishers:
/ydlidar_ydlidar_ros2_driver_node: /scan
Subscribers:
/rviz_1773000000_4294967295: /scan
Reliability: Best Effort
Durability: Volatile
Filter: 0
Frames:
laser_frame: OK
Time:
Header stamp: 1773000000.000000000
Header frame_id: laser_frame
Ranges:
size: 720
min: 0.010000
max: 64.000000
mean: 0.000000
first: 0.000000
last: 0.000000
Intensities:
size: 720
min: 0.000000
max: 0.000000
mean: 0.000000
first: 0.000000
last: 0.000000
Reset
Publishing Status
Fixed Frame [drop_frames]
Publishing Status
Time [drop_frames]
Publishing Status
Transformation [drop_frames]
Publishing Status
Message [drop_frames]
Publishing Status
Filter